

Quiz (Carolina Reaper)

Q1. Convert $x^2 + 6x + 8$ to vertex form.

- (A) $(x + 3)^2 - 1$
- (B) $(x + 4)^2 + 4$
- (C) $(x + 3)^2 + 1$
- (D) $(x + 2)^2 - 4$
- (E) $(x + 4)^2 - 8$

Q2. Complete the square for $x^2 + 4x$.

- (A) $(x + 2)^2 - 4$
- (B) $(x + 2)^2 + 2$
- (C) $(x + 1)^2 - 1$
- (D) $(x - 2)^2 - 4$
- (E) $(x + 1)^2 + 1$

Q3. What is the vertex of the parabola $x^2 + 2x - 6$?

- (A) $(-1, -7)$
- (B) $(-2, 6)$
- (C) $(-1, -5)$
- (D) $(-2, -2)$
- (E) $(1, 5)$

Q4. Convert $x^2 - 8x + 12$ to vertex form.

- (A) $(x - 4)^2 - 4$
- (B) $(x + 4)^2 - 4$
- (C) $(x - 2)^2 - 8$
- (D) $(x - 4)^2 + 4$
- (E) $(x + 2)^2 - 8$

Q5. Complete the square for $x^2 - 6x$.

- (A) $(x - 3)^2 - 9$
- (B) $(x + 3)^2 - 9$
- (C) $(x + 3)^2 + 3$
- (D) $(x - 3)^2 + 3$
- (E) $(x - 2)^2 + 2$

Q6. What is the equation of the parabola with vertex $(-2, 3)$ and x -intercept $(1, 0)$?

- (A) $y = (x + 2)^2 + 3$
- (B) $y = (x - 2)^2 - 3$
- (C) $y = -(x + 2)^2 + 3$
- (D) $y = (x + 1)(x - 3)$
- (E) $y = (x - 1)(x + 3)$

Q7. Convert $x^2 + 10x + 24$ to vertex form.

- (A) $(x + 5)^2 - 1$
- (B) $(x + 4)^2 - 8$
- (C) $(x + 5)^2 - 1$
- (D) $(x + 6)^2 - 12$
- (E) $(x + 4)^2 - 16$

Q8. Complete the square for $x^2 + 2x - 3$.

- (A) $(x + 1)^2 - 4$
- (B) $(x + 2)^2 + 2$
- (C) $(x + 1)^2 - 2$
- (D) $(x - 1)^2 - 2$
- (E) $(x + 1)^2 + 1$

Open Ended Questions (FRQ)**Q9.** Prove that the vertex form of $ax^2 + bx + c$ is $a(x - h)^2 + k$, where (h, k) is the vertex of the parabola.**Q10.** Find the vertex and x -intercepts of the parabola $y = x^2 - 4x - 5$. Explain the steps of your solution and provide supporting work.**Chef's Tips**

- Emphasize simplifying expressions to obtain the vertex form. 1
- Caution students to ensure the completion of the square is done correctly by checking that the original expression equals the transformed expression.
- Remind students to clearly identify the coordinates of the vertex for transformed equations.